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**University College Cork, Ireland**  
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**Title: Life cycle assessment of animal-based foods and plant-based protein-rich alternatives:  
An environmental perspective**

**(Short title: Environmental life cycle assessment of animal-based food and plant-based alternatives)**

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## Abstract

### Background

In the European Union proteins for food are largely animal-based consisting of meat and dairy products. Almost all soy but also a larger part of pulses and cereals consumed in the European Union are used for animal nutrition. While livestock is an important source of proteins, it also creates substantial environmental impacts. The food and feed system is closely linked to the planetary and health boundaries and a transformation to healthy diets will require substantial dietary shifts towards healthy foods, such as nuts, fruits, vegetables, and legumes.

### Results

Extruded vegetable meat alternatives consisting of protein combined with amaranth or buckwheat flour and a vegetable milk alternative made from lentil proteins were shown to have the potential to generate significantly less environmental impacts than their animal-based counterparts in most of the environmental indicators examined, taking into account both functional units (mass and protein content). The underlying field-to-fork LCA models include several variants for both plant and animal foods. The optimised plant-based foods show a clear potential for improvement in the environmental footprints.

### Conclusions

Development of higher processed and therefore higher performing products is crucial for appealing to potential user groups beyond dedicated vegetarians and vegans and ultimately achieving market expansion. The Protein2Food project showed that prototypes made from

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Europe-grown legumes and pseudo-cereals are a valuable source for high-quality protein foods and despite being substantially processed they could help reduce the environmental impact of food consumption.

**Keywords:** Protein transition, protein-rich food, plant-based meat substitutes, plant-based milk substitutes, life-cycle assessment, sustainable food

## Introduction

The world's population could grow up to more than 9 billion by 2050 giving rise to a largely increased demand for food in general and protein food and feed in particular<sup>1</sup>. In addition, increased demand for protein globally is driven by socio-economic changes such as rising incomes or increased urbanisation while recognizing the role of protein in a healthy diet<sup>2</sup>. Taking into account that protein undernourishment exists in some poorer populations in developing countries and an additional need of up to 50 % of protein food might occur by 2050<sup>3</sup>. Without changes in current standard diet patterns this will result in increased demand for animal-based protein and the need to produce more animal feed<sup>3,4</sup>. In fact, in the European Union (EU), proteins for food are largely animal-based, consisting of meat and dairy products. At global level, livestock provides a quarter, in the EU even more than 50 % of all the protein consumed in food (Table 1).

Table 1: Share of protein intake at EU-28 average<sup>5,6</sup>

At the same time the EU has a 70 % deficit in protein-rich grains that is met primarily by imports of GMO soya for concentrate feed from the USA and South America<sup>7,8</sup>. Almost all soy, but also a larger part of pulses and cereals consumed in the EU are used for animal nutrition (Table 2). While arable land accounts for about 60 % of EU-28 total agricultural land, domestic production of protein crops in the EU occupies about 1.5 % and that of soy beans about 1 % of total arable land<sup>9,10</sup>. In contrast, 68 % of the total agricultural land is used for animal production<sup>11</sup>.

Table 2: Crop use for food and feed in the EU-28<sup>5,6</sup>

Although livestock farming is now an important source of protein, it also causes considerable environmental pollution<sup>12–14</sup>. Agriculture is responsible for about 10 % of the EU's GHG emissions and nearly 70 % of those come from the animal sector, followed by dairy products<sup>11,15</sup>. This contribution to livestock GHG emissions is due to methane from enteric fermentation. In addition, livestock farming is responsible for, among others, nitrous oxide and ammonia emissions to air as well as nitrogen and phosphorus pollution from fertilizer and manure used on pasture and croplands occupied for fodder production and manure management<sup>12,16,17</sup>. Furthermore, intensive livestock farming is estimated to account for 78 % of agriculture's negative impact on biodiversity<sup>1</sup>.

The food and feed systems are therefore closely linked to the planetary and health boundaries. Animal-based foods are a major driver of boundary transgressions<sup>18</sup>. The concept of planetary boundaries has been first published by Rockström et al.<sup>19</sup> and was further developed by Steffen et al.<sup>20</sup>. It identifies nine processes that regulate the stability and resilience of the Earth system and it defines quantitative boundaries the crossing of which increases the risk of generating large-scale abrupt or irreversible environmental changes.

The planetary boundaries framework has recently been applied to set guardrails of diets entailing food consumption patterns which would safeguard both, human as well as planetary

health boundaries<sup>21</sup>. Three key areas, namely climate change, phosphorus cycling and biodiversity loss are already exceeded through today's food consumption, while nitrogen cycling is close to the upper tolerance level. Maintaining existing food patterns with a growing world population until 2050 will increase the transgression levels and exceed the boundaries of cropland use<sup>18</sup>. Animal-based foods also contribute to a large transgression of healthy food consumption on a global average and even more when considering an average "northern" diet<sup>21</sup>. In the "northern" diet this is caused in particular by consumption of red and white meats, dairy and eggs, and by tuberous vegetables with high starch content<sup>18</sup>. Transformation to healthy diets by 2050 will require substantial dietary shifts, including a greater than 50 % reduction in global consumption of unhealthy foods, such as red meat and sugar, and a greater than 100 % increase in consumption of healthy foods, such as nuts, fruits, vegetables, and legumes<sup>18,21</sup>.

The work presented here was carried out within the EU-Horizon 2020 project PROTEIN2FOOD (AMD-635727). The project aimed at developing prototypes of innovative, nutritious, and protein-rich plant-based food, as an alternative to conventional animal-based foods such as pig, poultry and beef meat as well as cow milk. Studies on the environmental impact of vegetable alternatives and animal-based food show that vegetable proteins perform better than comparable animal products in terms of GHG emissions and other impact categories such as eutrophication or acidification<sup>22–25</sup>.

The prototypes within the Protein2food project were made from Europe-grown grain legumes such as lentils, faba beans and lupins, combined with pseudocereals, such as buckwheat, quinoa and amaranth. While buckwheat is currently stably domestic; the EU quinoa and amaranth production is increasing and introduced around Europe<sup>26</sup>. The prototypes developed in the project were vegetable meat and cow milk replacers to support the protein transition by reducing animal-based protein consumption and dependency on protein import to the EU. These products have high consumer acceptability and are particularly promising for the fulfilment of human dietary requirements due to their well-balanced amino acid profiles<sup>27</sup>.

This article examines the sustainability of an extruded vegetable meat alternative („VMA-extrudate“) providing an alternative for chicken meat and a vegetable milk providing an alternative for cow's milk. The VMA extrudate developed in Protein2Food is a high-moisture meat substitute without soy or wheat by using grain legumes and pseudocereals. It can be produced in form of nuggets, chunks, strips and burgers and can be flavoured with various aromas to satisfy different tastes. The VMA products are characterised by well-defined fibre formations that closely resemble meat structures and have enhanced taste sensation. The high-protein milk substitute is based on lentil protein isolate and scored as high as commercial beverages (e.g. soy, oat-based) in sensory tests conducted as part of the Protein2Food project<sup>28</sup>. Unlike many other plant-based beverages, it contains as much protein as cow's milk and is allergen free. Therefore, it could serve as a high protein alternative to soymilk.

The Protein2Food prototypes are highly processed foods. One of the guiding questions in the project was, whether these innovative plant-based foods are really more environmentally sustainable than the traditional animal-based counterparts. This question was addressed by comparing the environmental footprints of the plant-based protein-rich prototypes against animal-based references.

## 1. Materials and Methods

Environmental profiles of the selected food products have been calculated through life cycle assessment following the ISO 14040 and ISO 14044 standards as well as taking into account

guidance provided by the ILCD methodology<sup>29</sup> and the World Food LCA Database<sup>30</sup>. The product variants presented in this article are VMA extrudates using lupin as protein source and amaranth or buckwheat flour as well as vegetable milk made from lentil proteins (Table 3). An “optimised” variant has been included for both plant-based products to further explore improvement potentials. Lupins and particularly lentils are legumes with relatively low yields (see Table 5); the optimised variants simulate a situation that might be achievable if producing the same prototypes with higher yielding protein crops such as faba beans. On top of that, using faba beans instead of lupins would also save a de-oiling step during grain processing. Table 4 shows the nutritional composition of the examined products as well as their water content.

Table 3: Product variants examined

Table 4: Nutritional composition of product variants examined

The animal-based products were modelled as a “high performance” and a “conventional” variant. “High performance” here refers mainly to animal varieties with a high food conversion rate (FCR) which translates into animals with a high output of meat or milk respectively per amount of feed. “Conventional” refers to varieties with a medium food conversion rate. For instance, in the case of chicken FCR of 1.7. (high performance) and 2.2 (conventional) were assumed. A FCR of 2.2 means that 2.2 kg of feed is required per 1 kg of chicken live weight. This differentiation was made to benchmark the Protein2Food prototypes against a realistic range of corresponding animal-based products sold on the market.

The LCA was designed as a ‘cradle-to-gate’ LCA comprising all the life cycle steps from biotic and abiotic raw material sourcing up to the final food products at the factory gate; including all transports as well as all energy and raw material pre-chains. Production and disposal of the infrastructure (machines, transport media, roads, etc.) and their maintenance (spare parts, heating of production halls) as well as packaging material related to the final products were excluded based on Agrifootprint 2.0<sup>31</sup>.

In terms of the geographic scope, the LCA focuses on supplying the European market (EU28). This means that all life cycle steps, with the exception of imported soybeans/soybean meal (Brazil) and palm oil (Malaysia), take place within Europe. For example, an average European dataset for electricity generation were used for crop processing, husbandry and food production.

Figure 1 and figure 2 provide schematic flow charts illustrating the system boundaries and the main value chain steps using the example of chicken meat and VMA extrudate. These boundaries were considered to reflect the main differences between the life cycle of Protein2Food prototypes and animal-based products.

Figure 1: Process flow chart of a Protein2Food meat substitute prototype

Figure 2: Process flow chart of a traditional fibre meat (chicken) product

The functional unit (FU) is a key element of an LCA. It enables comparison between environmental performances of different systems or options (here: protein foods) on an equivalent basis. In Protein2Food, it was decided that protein content would be the adequate FU given the overall objective of developing plant-based food prototypes that are rich in proteins

and suitable to replace animal-based proteins also from a nutritional standpoint. In addition, supplying the consumer with proteins is the most recognized function of animal-based products<sup>25</sup>.

On the other hand, many consumers simply buy a portioned food item offered on the shelf or purchase a defined amount at the shop counter. While it does not reflect specific nutritional aspects of the food it could be assumed that nutritional requirements are satisfied within the overall diet<sup>32,33</sup>. Also, mass is the most common FU in LCAs of food products<sup>25,33</sup>. As a sensitivity check, a mass-based and a protein-based FU, were applied in the LCA, as well as an energy-based FU in the case of milk.

The core inventory data collected and applied are shortly described in the following. A table overview of key inventory data sources is also provided in the electronic supplementary material Table S1.

**Crop cultivation data and drying:** The agricultural production of protein-rich crops as well as feed crops takes place at many different locations throughout the EU with main producer countries varying for each crop. In the context of Protein2Food, a crop model was developed such as to reflect as much as possible the condition of major geographic regions with high production volumes of the individual crops. Table 5 shows the crops relevant to Protein2Food and the yields assumed in the LCA calculations ("model yield"). The table furthermore shows the three main producer countries within the EU that are taken into account and the yield ranges across Europe as provided by FAO or Eurostat respectively. Table 6 summarizes the key cultivation parameters for the Protein2Food crops. Main data sources are primary data from field trials (field data) obtained from Protein2Food project partners as well as secondary data sources, such as crop-specific literature (see table 6). Nitrate leaching is calculated based on the SQCB-model (Sustainable quick check for biofuels) from Faist et al.<sup>34</sup>, taking into account soil clay content, rooting depth, N fertilisation, N uptake by crop/residues and nitrogen in soil organic matter. Emission factors for ammonia emission from fertilizer application have been taken from Müller-Lindenlauf<sup>35</sup> and nitrous oxide emissions from fertilizer application and crop residues remaining on the field are calculated according to the calculation rules of IPCC<sup>36</sup>. Direct land use change (dLUC) was calculated with a crop-specific approach using the 20-year time horizon as recommended by PAS 2050<sup>37</sup> and based on statistical data published by FAO on land use change from grassland and forest area to agricultural land in the respective import countries. The drying of the crops is assumed to take place relatively close to the cultivation areas.

Table 5: Main cultivation countries and yield ranges within the EU<sup>38,39</sup>

Table 6: Key Cultivation parameters of Protein2Food crops<sup>26,40–44</sup>

**Crop processing data:** Conventional processing of agricultural products into feed is usually operated at industrial scale conditions. Inventory data on processing of Protein2Food crops into protein isolates and flours respectively were available at laboratory or pilot scale collected by project partners. For the purpose of a "fair" comparison of those data, they were scaled-up to reflect commercial scale operations by adjustment of process efficiencies regarding protein and flour yields as well as energy efficiency. The processing stages for Protein2Food crops into protein isolates and flours and the respective CED results are listed in the electronic supplementary material Table S2. The processing of crops into feed is assumed to take place in the region of the cultivation areas, except for imported crops that are processed partly in the cultivation country and partly in Europe.



**Protein foods production data:** The inventories of chicken husbandry and slaughterhouse operations were taken from<sup>31,45–47</sup>. The inventory of milk cow husbandry and milk production were taken from Agrifootprint 2.0<sup>31</sup>. Inventory data on Protein2Food protein-rich extrudate as well as vegetable milk were available at laboratory or pilot scale collected by project partners. Those data were also scaled-up to reflect commercial scale operations by improved intermediate product yields and energy efficiencies. The respective CED results are listed in the electronic supplementary material Table S2. The food production represents typical European conditions and therefore, it is located all over Europe. It is assumed that animal husbandry and slaughtering or milk production as well as various processing steps for the plant products are located at different production sites. For the truck transports to the drying, harvest processing and food production steps, the model assumes a total transport distance of 950 km, 750 km and 850 km for chicken meat, milk and the plant products, respectively, with the exception of the imported crops, where the overseas transports are added.

**Environmental Impact Assessment:** Environmental footprints were generated by application of a set of environmental categories with related category indicators and characterization models suitable to address areas of environmental concern of the European agriculture and food sector<sup>48</sup> (Table 7). Characterization models were also selected with respect to the avoidance of uncertainty in characterization factors and to the completeness and availability of inventory data. The choice is also based on the German Federal Environmental Agency (UBA) approach 2016, where the available methods have been discussed and evaluated<sup>49</sup>. With regard to the toxicity categories, the UBA this study<sup>49</sup> shows that the modification of individual characterization factors by the uncertainty factors documented in Rosenbaum et al.<sup>50</sup> would lead to considerable differences in the results. Therefore, the toxicity categories are not used for the derivation of substantiated comparative statements.

Table 7: Selected categories and assigned inventory parameters as well as characterization models used in the LCA<sup>51–55</sup>

## 2. Results

The carbon footprints of Protein2Food prototypes and those of their animal-based counterparts are shown in Figures 3 and 4. The carbon footprints of VMA extrudates (Figure 3) refer to 100 g of food product with a protein content of 30 g. Comparisons that were conducted on a mass basis show values of 235 g of CO<sub>2</sub>-equivalents and 240 g of CO<sub>2</sub>-equivalents for VMA extrudates from lupin protein combined with amaranth or buckwheat flour, respectively. The carbon footprint of the optimised vegetable VMA made from faba-bean protein in combination with amaranth (Variant 3) is 130 g CO<sub>2</sub> equivalents per 100 g food product. The corresponding carbon footprints of 100 g of chicken meat are 232 g of CO<sub>2</sub>-equivalents and 342 of CO<sub>2</sub>-equivalents for chicken breast obtained from high performance chicken and from conventional chicken, respectively. Compared to high-performance chicken, the optimised variant has a significantly lower carbon footprint.

Interestingly, the carbon footprint of high performance chicken is in the range of the VMA extrudates, if compared on a mass basis (left part of Figure 3) but becomes somewhat larger if compared on a protein equivalence basis due to the smaller protein content of chicken breast as compared to the VMA extrudates (right part of Figure 3). “High performance chicken” are typically used in highly industrialized broiler husbandry, which often implies highly adverse



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effects on animal welfare. Yet, when additionally considering the greenhouse gas emissions caused by direct land use change associated with soy feed from overseas (here using the example of Brazil) the carbon footprint of chicken breast is almost double to threefold of that of the VMA extrudates if compared by protein content.

Figure 3: Carbon footprints of VMA extrudate versus chicken meat

In the case of cow's milk and lentil protein based milk (Figure 4), the differences are much more apparent. Even without accounting for direct land use change and when considering the energy based functional unit the carbon footprint of cow's milk is three times up to almost four times larger than that of the plant based milk. Soy feed is less relevant in cow's feed as compared to chicken feed and therefore greenhouse gas emissions associated with oversea soy feed has a smaller relative impact on the carbon footprint of cow's milk.

Figure 4: Carbon footprints of vegetable milk substitute versus cow's milk

Figures 5 and 6 show contributions of individual life cycle steps to the overall environmental profile of plant-based food products for each of the environmental indicators. Each pair of contribution bars represents the base product as well as its corresponding optimised variant product.

Figure 5: Contribution analysis: VMA extrudate

Figure 6: Contribution analysis: vegetable milk substitute

Contributions of processing (including processing of crops to intermediates, e.g. protein isolates, as well as production of food products) are clearly visible in most indicators. Share of processing is up to 75% (base VMA product, for non-renewable primary energy). Also high shares for processing are observed for most air-emission related indicators with the exception of ozone depletion, where nitrous oxide from farming is of high relevance. The burdens and resources from crop processing, which includes, for example, processes of drying and sorting, de-hulling and milling of seeds as well as the several stages of protein extraction, are to a good extent related to energy generation. With regard to photochemical ozone formation, the production of chemicals (hydrogen chloride and sodium hydroxide) and, in the case of lupine, hexane emissions from solvent extraction also play a significant role. Further indicators, where farming stage is the predominant contribution, are land use, phosphate rock and aquatic eutrophication (the latter due to nitrate leaching as a result of the nitrogen cycle in the soil and above/below-ground biomass). As protein isolation modelled here is an aqueous process, processing is also the predominant life cycle step regarding blue water demand.

Figures 7 and 8 show comparative environmental footprints using spider-diagrams entailing 12 environmental indicators. Each diagram displays two plant-based products (base product and optimised variant) along with their reference animal-based product. Net results of all products for each indicator are provided in the electronic supplementary material Table S3 and S4.

In the spider diagrams, the worse result per indicator and variant is set to 100 % and the relative better results are shown as percentages of it. The environmental footprint of plant-based VMA (base product) tends to be better than that of chicken meat for most of the indicators with the exception of cumulative energy and process water demand as well as land use. Strongly reduced nitrogen fertiliser requirements of legume crops relative to classical feed crops are the reason for favourable results of meat substitutes in aquatic eutrophication. On the other hand, both

cumulative energy and process water are associated with the several processing steps from crop to protein isolate, whereas land use is closely related to yields of protein crops. The results of the optimised variant of the plant-based VMA indicate that significant improvements in the ecological footprint of the VMA products can be achieved in all categories. As a result, the optimised variant here performs clearly better also regarding land use and process water demand than high-performance chicken.

Similarly, the environmental footprint of lentil-based milk is mostly favourable as compared to cow's milk, with land use being the single indicator showing a clear disadvantage. With regard to aquatic eutrophication, the comparative results of the milk substitute show a more differentiated picture. The lentil-based milk tends to have a lower impact when compared to the conventional cow's milk, but provides an ambiguous result when compared to the more efficient cow's milk. This is explained by the relatively low crop yield per ha for lentils and by the fact that nitrate leaching is a function of agricultural area. Thus, the more cropland needed to provide the feed or food ingredients per functional unit, the more nitrate emissions from the soil occur. The optimised variants show a clear advantage due to the higher yield of faba bean, especially in the two categories aquatic eutrophication and land use. The overall advantage of the plant-based prototype is more outstanding in the case of milk. However, both plant-based prototypes still bear a substantial improvement potential that can be seen in the optimised variants.

Figure 7: Environmental footprint in comparison: VMA extrudate

Figure 8: Environmental footprint in comparison: vegetable milk substitute

### 3. Discussion

At the time when the Protein2Food project started most meat replacers available on retail shelves were based on quorn (mycoprotein-based protein), wheat protein (gluten) and soy protein<sup>56</sup>. Recently also pea protein is increasingly being used in meat alternatives<sup>57</sup>.

Carbon footprints per 100 g meat replacers are reported to be 272 g CO<sub>2</sub>-equivalents if soy meal based, 130 g CO<sub>2</sub>-equivalents if tofu based and 390 g CO<sub>2</sub>-equivalents for minced soy<sup>25,58,59</sup>. 238 g CO<sub>2</sub>-equivalents (cradle to factory gate) was reported for 100 g of a meat replacer from isolated pea protein<sup>57</sup>. A meat replacer using potatoe and soy protein accounted for 347 g CO<sub>2</sub>-equivalents per 100 g (cradle to factory gate)<sup>60</sup> while wheat gluten based meat replacer was calculated to generate 381 g CO<sub>2</sub>-equivalents per 100 g<sup>29</sup>.

For the soy based meat replacers it is unclear from the information provided in the cited literature as to which extent they contain highly processed concentrated and/or isolated proteins. It also remains unclear whether direct land use change was considered in the cited carbon footprints of soy products. Overall, the lupin-based meat replacer prototypes developed within Protein2Food are well within the lower range of the carbon footprints reported for meat replacers based on soy (excluding the more mildly processed tofu) and pea proteins.

Plant-based milk alternatives can be made from five plant categories which are legume-based (example: soy, pea, lentils), cereal-based (example: oat, rice), pseudo-cereal-based (example: quinoa, amaranth, buckwheat), nut-based (example: almonds) and seed-based (example: hemp, linseed). Of the examples mentioned only the milk alternatives made from soy, pea and lentil proteins have protein contents comparable to that of cow's milk<sup>61</sup>.

Carbon footprints per 100 g of milk alternative are reported to range between 22 to 51 g CO<sub>2</sub>-equivalents for soy-based drinks<sup>58,62–64</sup> and 61 g CO<sub>2</sub>-equivalents for a pea-based drink<sup>65</sup>. The carbon footprint calculated for the lentil-based prototype developed within Protein2Food is at the lower end of the numbers reported for protein-rich milk alternatives.

The LCA results of Protein2Food prototypes rely on assumptions regarding process efficiencies along the value chain and the achievable economic value of by-products from crop processing. Yet, the assumptions were backed by expert judgments from participating SMEs amongst others, thereby limiting uncertainty substantially. In fact, the results presented here indicate that Protein2Food meat replacers and milk alternatives made from protein crops other than soy and pea may have similar environmental profiles as those.

It is also worthwhile to mention that from a food producer's perspective lentil proteins are promising raw materials for forming a potential future platform for the production of plant based dairy alternatives. On the other hand, from a farmer's perspective, beans are preferred because they are easier to grow, and have rather steady generally higher yield levels<sup>26</sup>. The use of faba beans as a further crop source is examined in the Protein2Food prototypes in form of LCA optimisation scenarios. The faba bean-based meat replacers thus also provide insight into variability in the agricultural phase, as they represent a high-yield legume crop as protein source. Hence results related to "variant 1" meat replacers and "variant 3" replacers do cover agricultural variability in terms of legume yields in a range from 0.8 t/ha to 3.5 t/ha (for lentil and faba bean, respectively, see also Table 6). Similar trends are expected for yield variations within the same crop.

Cultivation of faba beans in those optimised scenarios versus lentil and lupin thus here also represents sensitivity analysis regarding the agricultural production phase. With faba bean being a legume crop showing a rather high yield potential (within legume crops), it also reflects a rather "efficient" legume crop cultivation, e.g. in terms of land required per crop unit, overall fertilizer amount per crop unit. Results indicate that increasing of legume crop yields and use of high-yield legume crops as protein sources for plant-based meat substitutes and vegetable milk alternatives are very promising from an environmental perspective. The optimised meat substitute prototype (variant 3) in particular shows this potential, as it has a significantly lower carbon and further environmental footprint compared to high-performance chickens.

Overall, the optimised product variants indicate that the environmental footprints of Protein2Food prototypes could further benefit from legumes with high field yields.

#### 4. Conclusions

Global food production is the largest pressure caused by humans on Earth and it is clear that the transition to sustainable food systems will not happen without a shift in people's diets<sup>66</sup>. Despite both, the health and environmental benefits, changing consumer preferences towards a diet containing low/no animal food is extremely difficult<sup>67,68</sup> and the global average per capita rate of animal product consumption has continued to increase<sup>69,70</sup>.

In western countries though, vegetarianism and reduced meat diets are increasing and in recent years, plant-based meat substitutes have experienced unprecedented growth<sup>71,72</sup>. New processing techniques have allowed the development of better performing products in terms of taste and texture and the use of a broader range of raw materials<sup>73</sup>.

Plant-based proteins already play an important role to satisfy future protein demand and overall, cereal proteins account for most of the dietary plant protein intake globally<sup>15</sup>. But food products obtained by using a blend of cereal and pulses can balance the amino acids and may improve the nutritive value of plant based protein foods<sup>3</sup>.

The Protein2Food project showed that prototypes made from Europe-grown legumes and pseudocereals are a valuable source for high-quality protein foods and despite being substantially processed such food could help reduce the environmental impact of food consumption, especially when replacing cow-milk-based food products. When replacing chicken, future aim should be to further reduce energy demand of the processing stage (as commercial production gets more and more mature) as well as optimize legume farming towards higher and stable yields. This confirms, that increasing the proportion of vegetable proteins and decreasing those of animal origin in the human diet is a win-win situation from both environmental and nutritional standpoints<sup>8</sup>. Future research should also be directed towards yield variations in legume farming, also depending on the type of crop management (organic versus conventional, tillage practices), the degree of irrigation and regarding its implications for water & energy requirements.

The need for increased consumption of legume-based proteins is challenged by the fact that the area of protein crops in Europe declined almost continuously over the last five decades, from 4.7 % of the arable area to about 1.5 % of the arable area<sup>26</sup> despite their agronomic and environmental benefits. A major underlying driver behind the reduction in the proportion of arable land used for protein crops is the increased comparative advantage in the production of starch-rich cereals in Europe over the production of protein-rich grain legumes<sup>68</sup>. Protein crops suffer from yield instability compared to cereals or rapeseeds, and yield fluctuations are one of the main reasons farmers give for not growing these crops<sup>26</sup>. The latter is a major obstacle in further expansion that urgently needs to be overcome and should be a main target for improving protein crop production. It would be important here that a further expansion of the area under protein crops does not lead to direct or indirect land use changes or potential water scarcity<sup>48</sup>, but to a diversification of crop rotations.

Development of higher processed and therefore higher performing products is crucial for appealing to potential user groups beyond dedicated vegetarians and vegans and ultimately achieving market expansion<sup>72</sup>. However, due to the high contribution of food processing to the environmental footprint, even with optimised processing, the highly processed plant-based meat and milk replacer foods are expected to still have a higher environmental impact than less processed products. Therefore, mildly processed protein crops such as cooked grain legumes should also be examined regarding their environmental performance in future research in order to identify if those are a suitable permanent component of diets as sources of low-cost dietary proteins for human consumption with a low-impact environmental profile.

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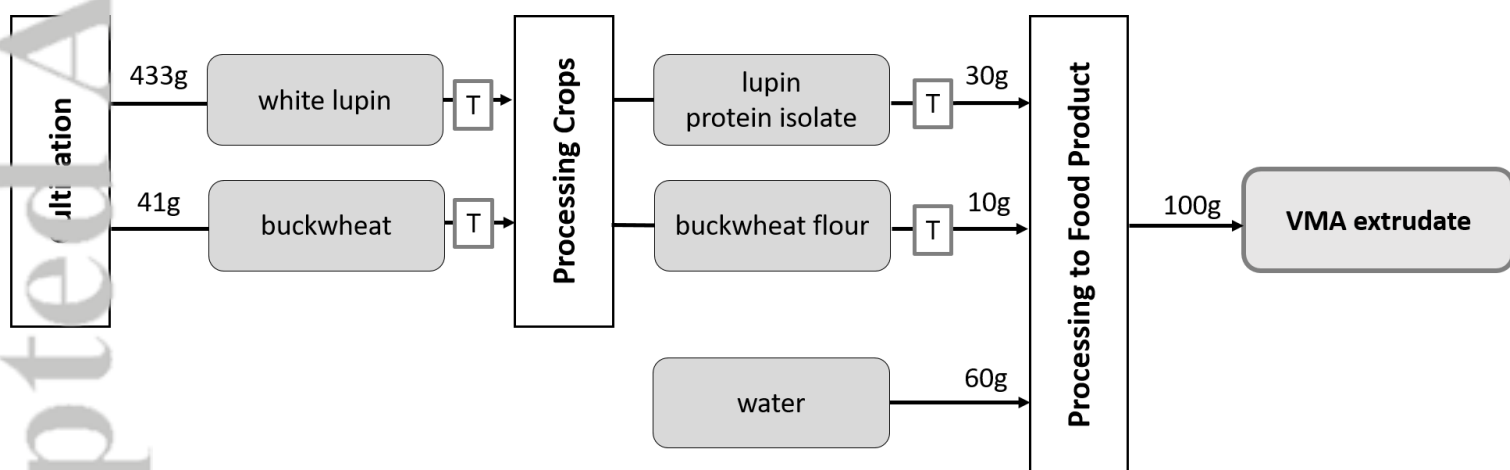


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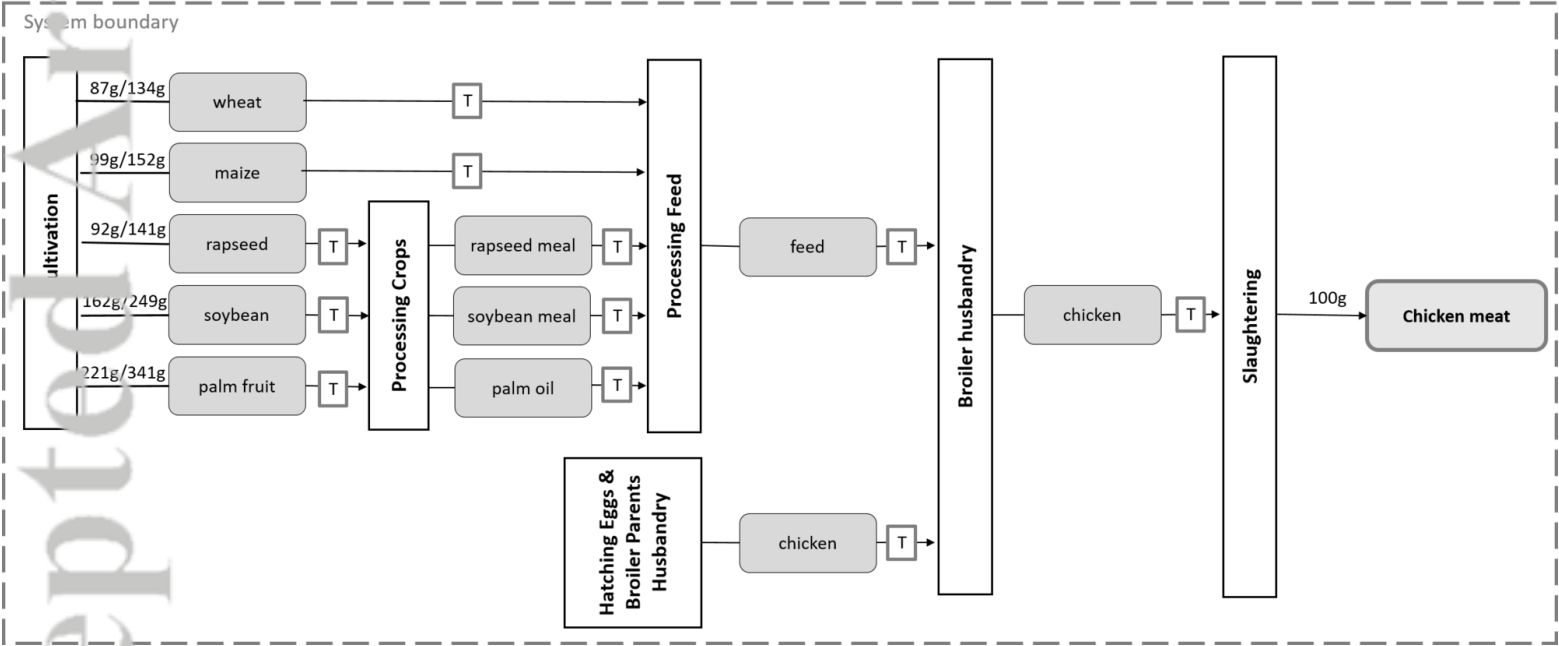
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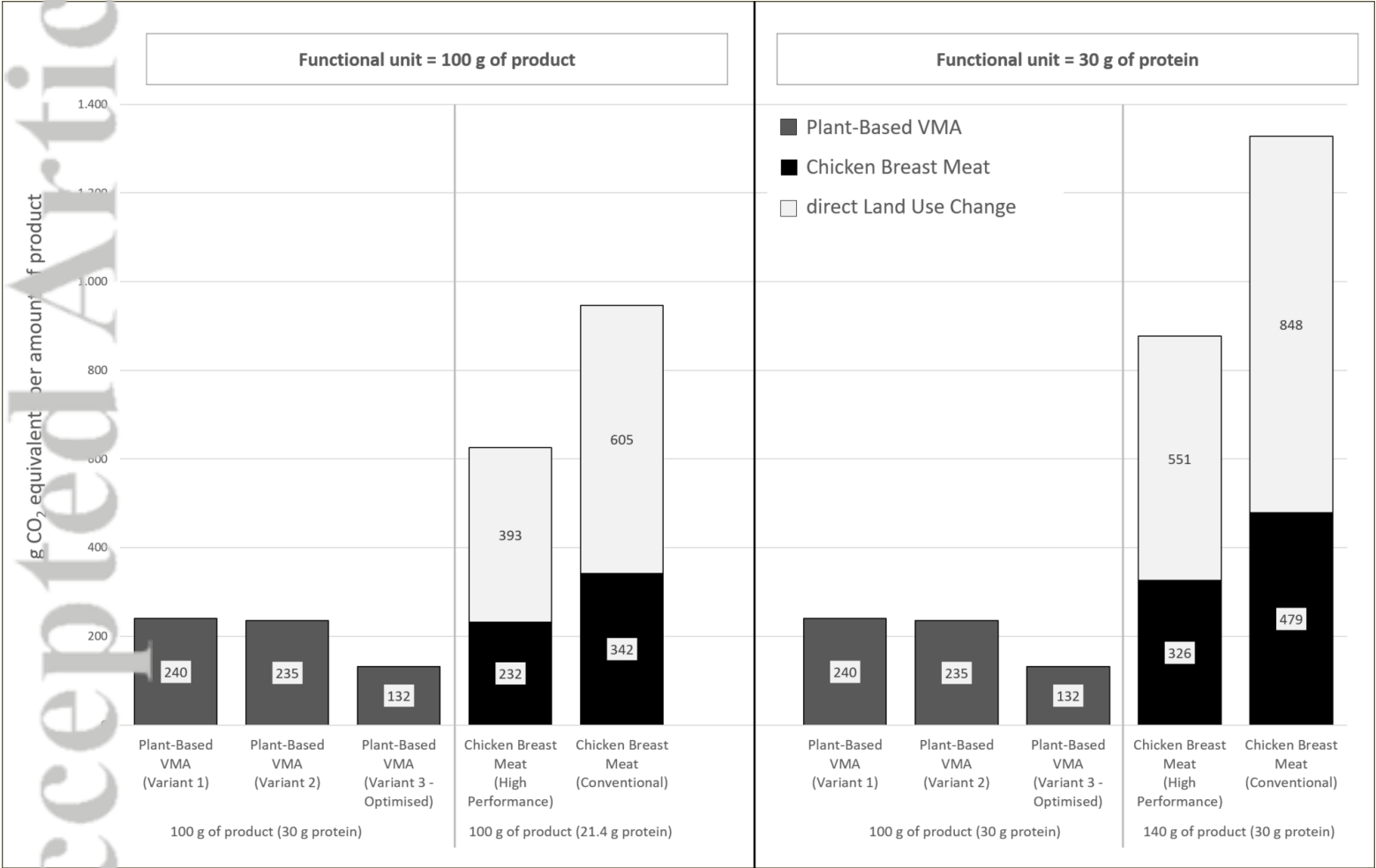
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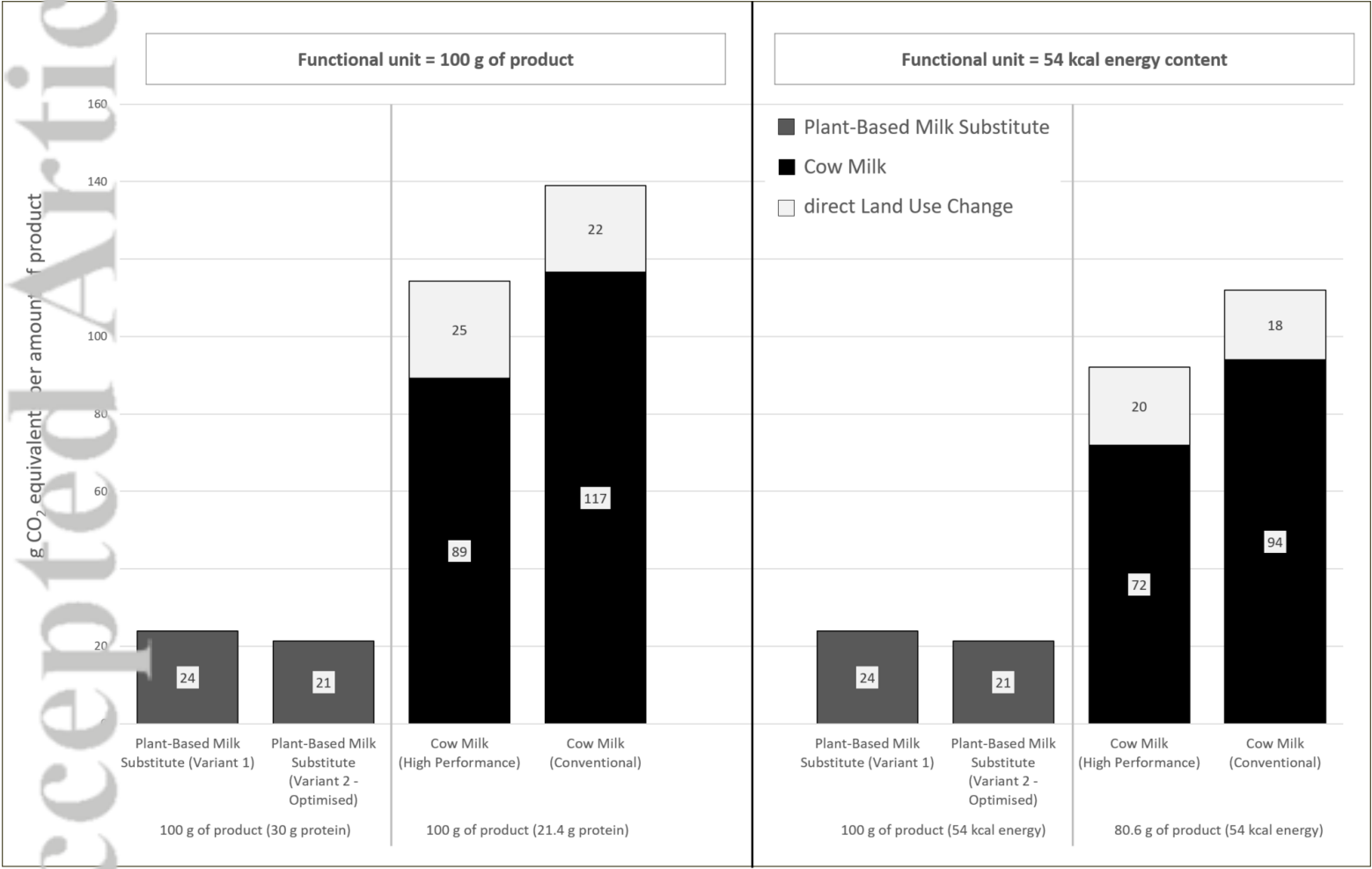
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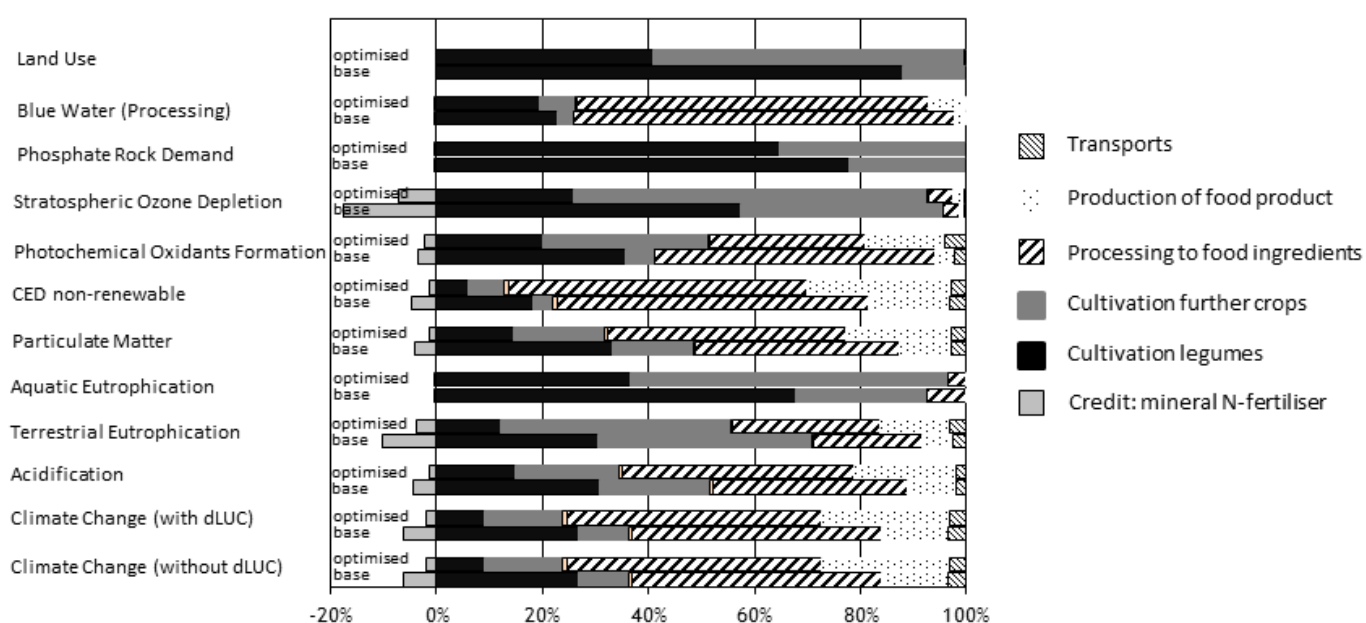


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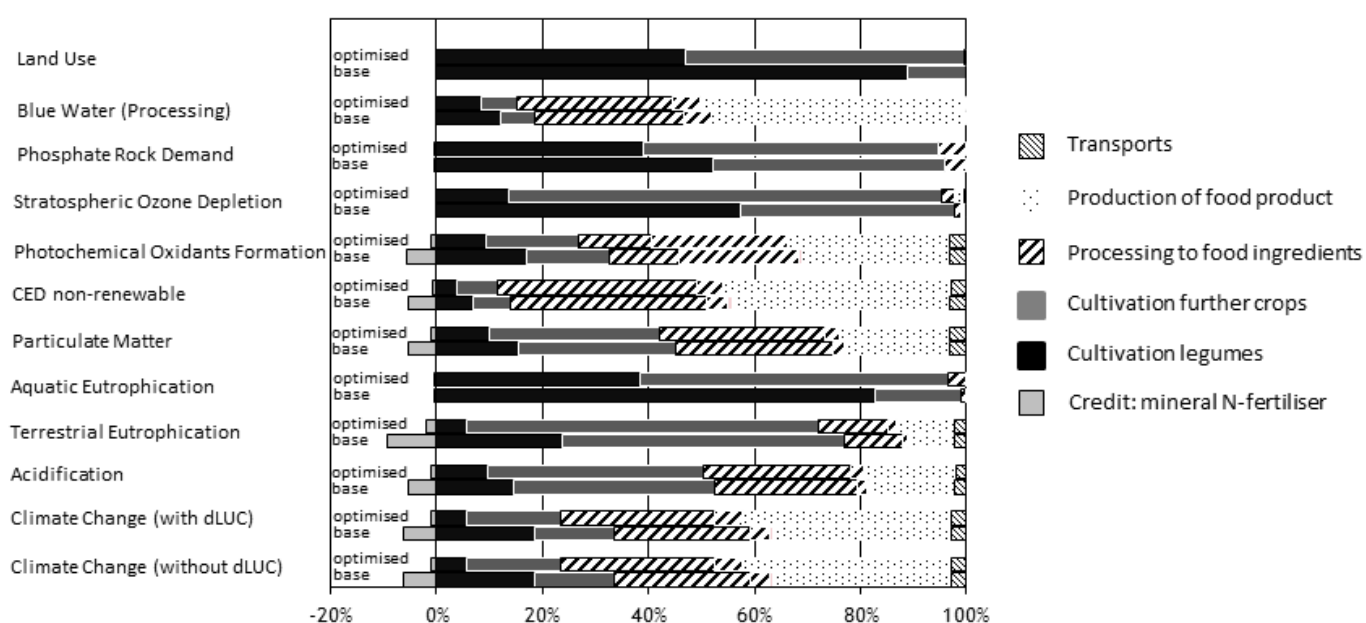


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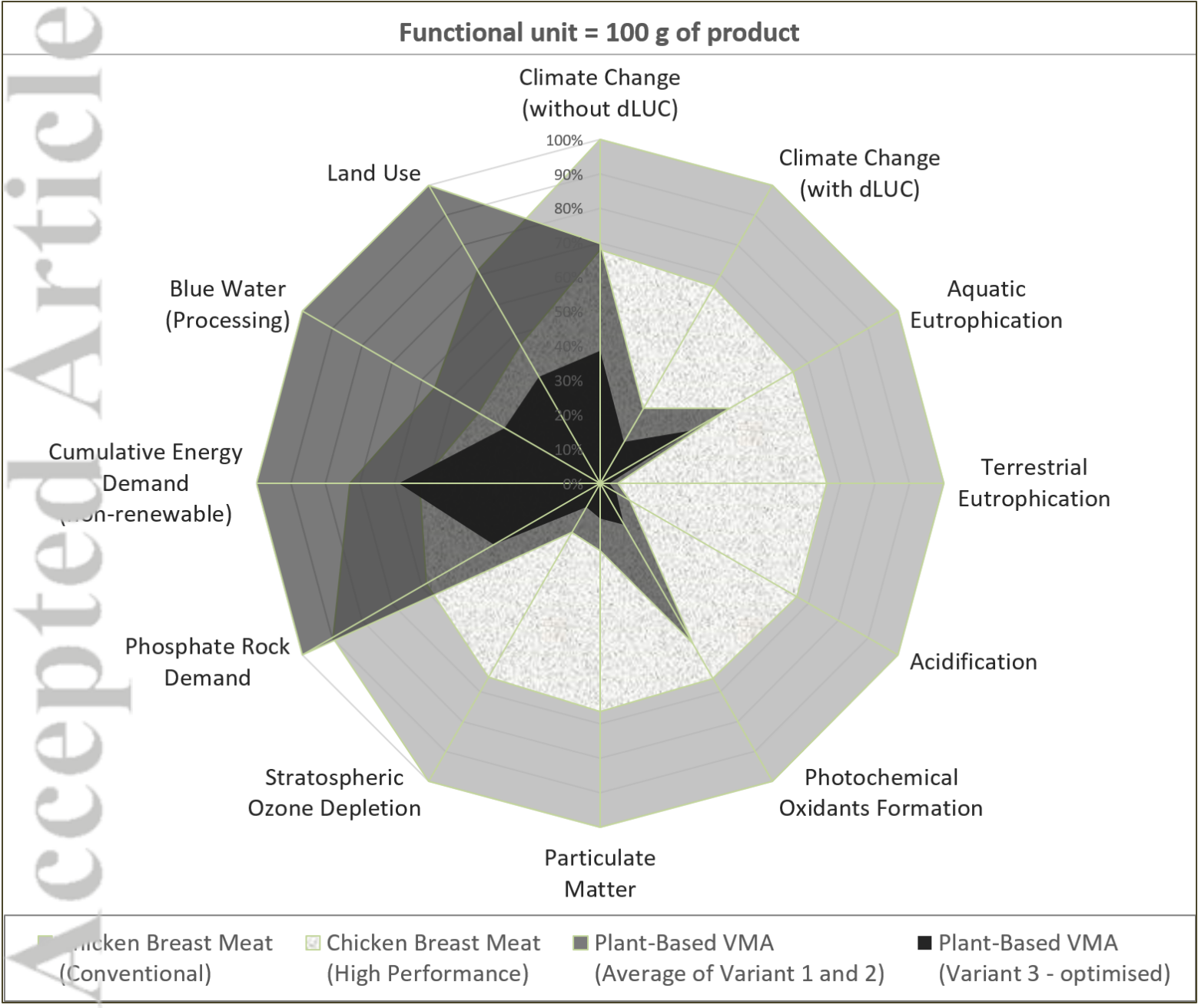




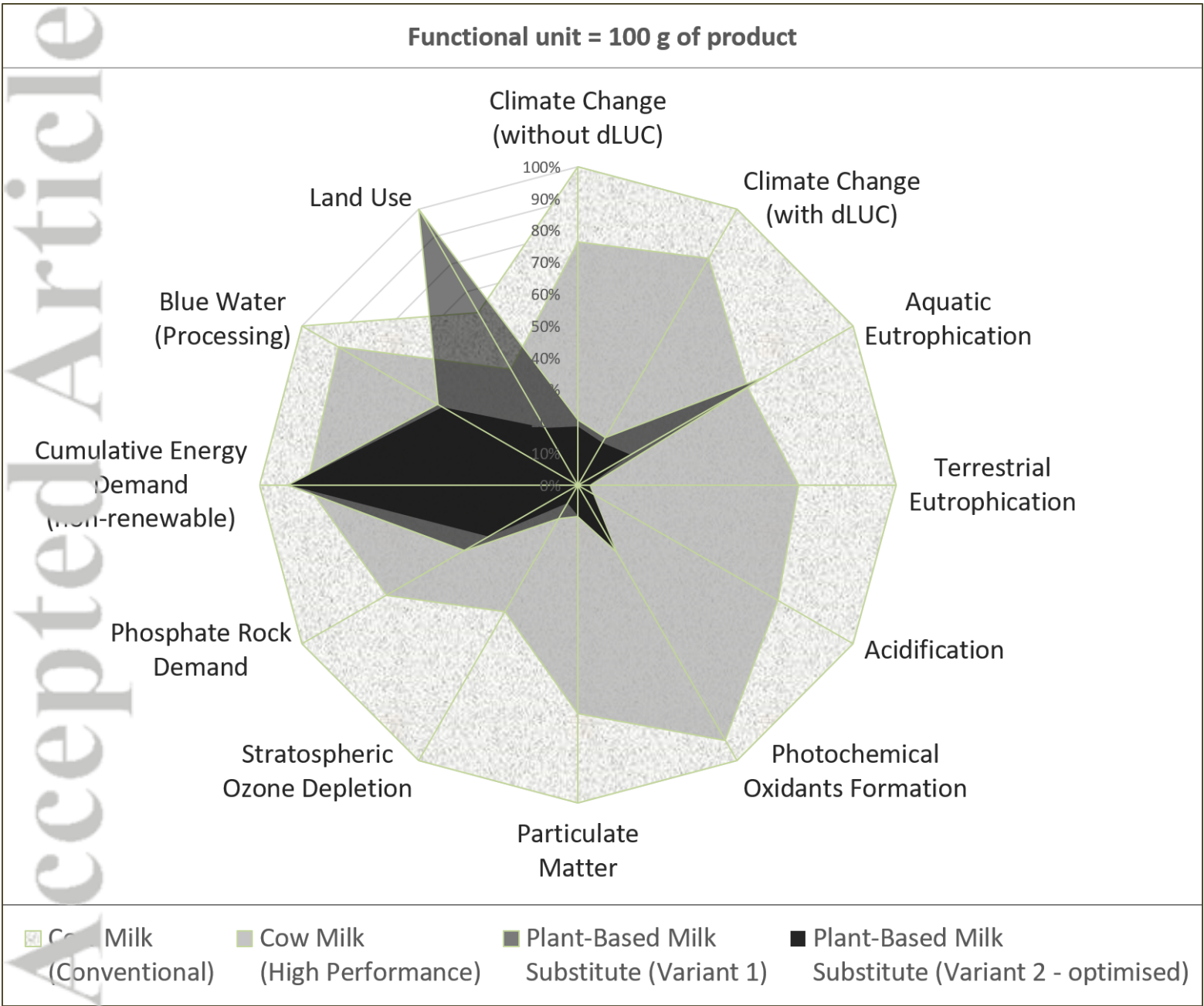
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Table 1: Share of protein intake at EU-28

|                          | (g/cap/day) | Share (%) |
|--------------------------|-------------|-----------|
| <b>Meat</b>              | 17.3        | 29        |
| <b>Dairy</b>             | 13.4        | 22        |
| <b>Eggs</b>              | 2.8         | 5         |
| <b>Fish, Seafood</b>     | 2.7         | 4         |
| <b>Cereals</b>           | 16.3        | 27        |
| <b>Starchy Roots</b>     | 1.7         | 3         |
| <b>Pulses</b>            | 1.5         | 2         |
| <b>Nuts + Oilcrops</b>   | 1.0         | 4         |
| <b>Vegetables+Fruits</b> | 3.0         | 5         |

Source: Own elaboration based on FAO<sup>5</sup> and Detzel et al.<sup>6</sup>

Table 2: Crop use for food and feed in the EU-28

|                | <b>Food</b>   | <b>Feed</b>   | <b>Share of<br/>Feed</b> |
|----------------|---------------|---------------|--------------------------|
|                | (Mio. Tonnes) | (Mio. Tonnes) | (%)                      |
| <b>Cereals</b> | 82.0          | 167.7         | 67                       |
| <b>Pulses</b>  | 1.5           | 2.1           | 58                       |
| <b>Soy</b>     | 2.2           | 30.3          | 93                       |

Source: Own elaboration based on FAO<sup>5</sup> and Detzel et al.<sup>6</sup>

Table 3: Product variants examined

| Protein2Food Prototype                                                                                                                                                                   | Animal-based reference                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| VMA with 30 % protein content                                                                                                                                                            | Chicken breast meat with 21.4 % protein content                                                |
| Variant 1: combination of lupine protein isolate and buckwheat flour<br>Variant 2: combination of lupine protein isolate and amaranth flour<br>Variant 3: optimised variant (simulation) | Variant 1: meat from “high performance chicken”<br>Variant 2: meat from “conventional chicken” |
| Vegetable milk substitute with 3.3 % protein content                                                                                                                                     | Cow milk with 3.3 % protein content                                                            |
| Variant 1: Lentil protein based milk substitute<br>Variant 2: optimised variant (simulation)                                                                                             | Variant 1: milk from “high performance cow”<br>Variant 2: milk from “conventional cow”         |



Table 4: Nutritional composition of product variants examined

| Product                               | Composition                                                           | Average nutrition values per 100 g wet mass |                       |                                |                           |                   |
|---------------------------------------|-----------------------------------------------------------------------|---------------------------------------------|-----------------------|--------------------------------|---------------------------|-------------------|
|                                       |                                                                       | Total energy (kcal)                         | Total fat content (g) | Total carbohydrate content (g) | Total protein content (g) | Water content (g) |
| Chicken breast meat                   | Chicken meat                                                          | 119                                         | 3.1                   | 0                              | 21.4                      | 76                |
| VMA (variant 1 and 2)                 | Lupin protein isolate in combination with amaranth or buckwheat flour | 136                                         | 1.0                   | 5.0                            | 30.0                      | 60                |
| VMA: optimised (variant 3)            | Faba-bean protein isolate in combination with amaranth flour          | 136                                         | 1.0                   | 5.0                            | 30.0                      | 60                |
| Cow milk                              | Cow milk                                                              | 67                                          | 3.8                   | 4.8                            | 3.3                       | 88                |
| Vegetable milk (variant 1)            | Lentil protein isolate, oil, sugar                                    | 52                                          | 3.3                   | 1.8                            | 3.3                       | 92                |
| Vegetable milk: optimised (variant 2) | Faba-bean protein isolate, oil, sugar                                 | 52                                          | 3.3                   | 1.8                            | 3.3                       | 92                |

Table 5: Main cultivation countries and yield ranges within the EU

| Crops                         | Model yield | Main cultivation areas within EU |           |                              | Yield range |
|-------------------------------|-------------|----------------------------------|-----------|------------------------------|-------------|
|                               | (t/ha)      | 1                                | 2         | 3                            | (t WM*/ha)  |
| <b>PROTEIN2FOOD Crops</b>     |             |                                  |           |                              |             |
| <b>Amaranth</b>               | 2.01        | n.a.                             | n.a.      | n.a.                         | n.a.        |
| <b>Buckwheat</b>              | 1.01        | Poland                           | Lithuania | France                       | 1.3-3.7     |
| <b>(White) Lupin</b>          | 2.51        | Poland                           | Germany   | France                       | 1.6-1.7     |
| <b>Lentil</b>                 | 0.81        | Spain                            | France    | Bulgaria                     | 0.6-1.4     |
| <b>Faba Bean</b>              | 3.51        | France                           | UK        | Italy                        | 3.2-3.8     |
| <b>OTHERS (EU crops)</b>      |             |                                  |           |                              |             |
| <b>Wheat</b>                  | 7.90        | France                           | Germany   | Poland<br>(UK by mass)       | 6.9-8.0     |
| <b>Maize</b>                  | 9.80        | France                           | Romania   | Hungary<br>(Germany by mass) | 3.8-8.7     |
| <b>Rapeseed</b>               | 3.40        | France                           | Germany   | Poland                       | 3.0-3.5     |
| <b>Sunflower</b>              | 3.02        | Romania                          | Bulgaria  | Spain<br>(Hungary by mass)   | 2.0-2.3     |
| <b>Spring Pea</b>             | 3.52        | France                           | UK        | Hungary                      | 4.5-7.4     |
| <b>Sugar Beet</b>             | 60.02       | France                           | Germany   | Poland                       | 73-87.3     |
| <b>Rye</b>                    | 5.01        | Germany                          | Poland    | Denmark                      | 2.9-5.8     |
| <b>OTHERS (oversea crops)</b> |             |                                  |           |                              |             |
| <b>Soybean (feed)</b>         | 2.82        | n.a.                             | n.a.      | n.a.                         | n.a.        |
| <b>Oil Palm</b>               | 192         | n.a.                             | n.a.      | n.a.                         | n.a.        |

Source: Own elaboration based FAO<sup>38</sup> and EC<sup>39</sup>

**Table 6.** Key Cultivation parameters of Protein2Food crops

| Key cultivation parameters                                | Unit                                  | Lentil | White lupin | Faba bean | Amaranth | Buckwheat | Data source <sup>(a)</sup>                                                              |
|-----------------------------------------------------------|---------------------------------------|--------|-------------|-----------|----------|-----------|-----------------------------------------------------------------------------------------|
| Diesel from sowing up to harvest                          | L / ha                                | 12     | 84          | 53        | 75       | 84        | Field data                                                                              |
| Nitrogen fertilizer                                       | kg N/ ha                              | -      | -           | -         | 120      | 20        | Field data; Kool et al. <sup>40</sup>                                                   |
| Potassium fertilizer                                      | kg K <sub>2</sub> O / ha              | 19     | 90          | 140       | 20       | 20        | Kool et al. <sup>40</sup> ; KTBL <sup>41</sup> ; Mujica et al. <sup>42</sup>            |
| Magnesium fertilizer                                      | kg MgO / ha                           | 4.8    | 12.5        | 17.5      | -        | -         | KTBL <sup>41</sup>                                                                      |
| Phosphorus fertilize                                      | kg P <sub>2</sub> O <sub>5</sub> / ha | 9.6    | 32.5        | 52.5      | 70       | 20        | Kool et al. <sup>40</sup> ; KTBL <sup>41</sup> ; Mujica et al. <sup>42</sup>            |
| Pesticides                                                | kg active ingredient / ha             | 1.3    | 1.3         | 1.9       | 0        | 0         | Field data; KTBL <sup>41</sup> ; Agrifootprint <sup>31</sup>                            |
| Crop yield (refers to water content of crop after drying) | kg / ha                               | 800    | 2500        | 3500      | 1975     | 1026      | Kool et al. <sup>40</sup> ; Agrifootprint <sup>31</sup> ; Pulvento et al. <sup>26</sup> |
| Water content of crop at harvest                          | %                                     | 20     | 16          | 17        | 12       | 20        | Field data; KTBL <sup>41</sup> ; LTZ <sup>43</sup>                                      |
| Is the crop dried after harvest?                          | -                                     | yes    | yes         | yes       | no       | yes       | -                                                                                       |
| Water content of crop after drying                        | %                                     | 14     | 14          | 14        | 12       | 14        | KTBL <sup>41</sup> ; LTZ <sup>43</sup>                                                  |
| Nitrogen fixation                                         | kg N / ha                             | 91     | 160-180     | 175       | -        | -         | KTBL <sup>41</sup> ; Sulas et al. <sup>44</sup>                                         |

**Table 7.** Selected categories and assigned inventory parameters as well as characterisation models used in the LCA

|                                                                                                                            | Elementary flow (examples)                                                                                                                                                                | Unit                     | Characterisation model        |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------|
| <b>Emission-related categories</b>                                                                                         |                                                                                                                                                                                           |                          |                               |
| <b>Climate change</b>                                                                                                      | to air: CO <sub>2</sub> *, CH <sub>4</sub> ** , N <sub>2</sub> O, C <sub>2</sub> F <sub>2</sub> H <sub>4</sub> , CF <sub>4</sub> , CCl <sub>4</sub> , C <sub>2</sub> F <sub>6</sub> , R22 | kg CO <sub>2</sub> -e/fu | IPCC <sup>51</sup>            |
| <b>Stratospheric ozone depletion</b>                                                                                       | to air: CFC-11, N <sub>2</sub> O, HBFC-123, HCFC-22, Halon-1211, Methyl Bromide, Methyl Chloride, Tetrachloromethane                                                                      | kg CFC-11-e/fu           | WMO <sup>52</sup>             |
| <b>Photo-oxidant formation</b>                                                                                             | to air: CH <sub>4</sub> , NMVOC, Benzene, Formaldehyde, Ethyl acetate, VOC, TOC, Ethanol                                                                                                  | kg O <sub>3</sub> -e/fu  | Carter <sup>53</sup>          |
| <b>Acidification</b>                                                                                                       | to air/water/soil: NO <sub>x</sub> , NH <sub>3</sub> , SO <sub>2</sub> , TRS***, HCl, H <sub>2</sub> S, HF                                                                                | kg SO <sub>2</sub> -e/fu | Heijungs et al. <sup>54</sup> |
| <b>Terrestrial eutrophication</b>                                                                                          | to air/soil: NO <sub>x</sub> , NH <sub>3</sub> , SO <sub>x</sub>                                                                                                                          | kg PO <sub>4</sub> -e/fu | Heijungs et al. <sup>54</sup> |
| <b>Aquatic eutrophication</b>                                                                                              | to water: COD, N, NH <sub>4</sub> <sup>+</sup> , NO <sub>3</sub> <sup>-</sup> , NO <sub>2</sub> <sup>-</sup> , P                                                                          | kg PO <sub>4</sub> -e/fu | Heijungs et al. <sup>54</sup> |
| <b>Particulate matter</b>                                                                                                  | to air: PM2.5, SO <sub>2</sub> , NO <sub>x</sub> , NH <sub>3</sub> , NMVOC                                                                                                                | kg PM2,5-e/fu            | De Leeuw <sup>55</sup>        |
| <b>Resource-related categories at inventory level</b>                                                                      |                                                                                                                                                                                           |                          |                               |
| <b>Land footprint</b>                                                                                                      | use of agricultural area                                                                                                                                                                  | m <sup>2</sup> *a/fu     | -                             |
| <b>Water footprint</b>                                                                                                     | blue water use                                                                                                                                                                            | m <sup>3</sup> /fu       | -                             |
| <b>Cumulated energy demand (CED, total)</b>                                                                                | hard coal, brown coal, crude oil, natural gas, uranium ore, hydro energy, solar energy, wind energy, biomass                                                                              | MJ/fu                    | -                             |
| <b>Cumulated energy demand (CED, non-renewable)</b>                                                                        | hard coal, brown coal, crude oil, natural gas, uranium ore                                                                                                                                | MJ/fu                    | -                             |
| <b>Cumulated energy demand (CED, renewable)</b>                                                                            | hydro energy, solar energy, wind energy, biomass                                                                                                                                          | MJ/fu                    | -                             |
| <b>Use of phosphorus</b>                                                                                                   | Phosphorus crude ore                                                                                                                                                                      | kg/fu                    | -                             |
| * CO <sub>2</sub> fossil und biogenic / ** CH <sub>4</sub> fossil und CH <sub>4</sub> biogenic / *** Total Reduced Sulphur |                                                                                                                                                                                           |                          |                               |